

Sunny Zhang

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EDUCATION

University of Waterloo, Faculty of Engineering

September 2026 – April 2031 (Expected)

Candidate for Bachelor of Applied Science in Mechatronics Engineering

- **Awards:** Recipient of Guo Family Entrance Scholarship (\$5,000) and President's Scholarship of Distinction (\$2,000)
- **Relevant Coursework:** Intro to Mechatronics, Digital Computation

SKILLS

Languages/Software: Java, Python, C++, HTML, CSS, JavaScript, Git, SolidWorks, AutoCAD, Onshape

Hardware/Power Tools: Arduino, Lathe, Milling Machine, Drill Press, Metal-Cutting Bandsaw, etc.

EXPERIENCE

Mechanical Engineering Intern – [Ace Manufacturing Inc.](#)

June 2026 – September 2026

- Executed **10+** design revisions adhering to customer requirements in **AutoCAD** for a burn-off oven facility addition
- Engineered **400+** liter pump enclosures with covers, **4m²** platforms with stairs, and **3m²** interlocking steel panel walls to integrate with existing infrastructure, enhancing safety and stability by conducting **shear stress calculations**
- Fabricated **20+** metal components and pipes for wastewater treatment systems using lathes, mills, bandsaws, etc.
- Performed inventory check for dimensions, types, and counts of all pipe fittings to streamline administrative quoting

PROJECTS

2D Mapping Vehicle

September 2026 – Present

- Engineered an **Arduino**-based **differential-drive** vehicle utilizing a servo-swept **time-of-flight sensor**, **quadrature wheel encoders**, and an **MPU6050 IMU** for 2D position tracking and mapping
- Designed a multi-layered chassis in **SolidWorks**
- Implemented **C++** embedded firmware in **PlatformIO**, fusing encoder odometry with gyro heading to process **differential-drive kinematics** for real-time coordinate tracking
- Developed a live **Python** plotting script to parse **serial data** and dynamically visualize the vehicle's mapped environment and detected boundaries
- Drafted **Fritzing** wiring diagrams for a dual-rail power system (**L298N** motor driver, **buck-regulated 5 V** logic rail), streamlining debugging and physical assembly

Professional Portfolio Website – [Onboarding Page](#)

September 2026 – Present

- Designed, styled, and deployed a personal website from scratch using **HTML**, **CSS**, and **JavaScript**, implementing a responsive layout and custom styling
- Built an interactive dark/light mode toggle using **JavaScript** to add dynamic functionality to the site
- Managed source code and version history using **Git**, hosting the repository on **GitHub** with a documented README
- Deployed the site live via **GitHub Pages**, making it publicly accessible

Robot Vacuum (Roomba)

April 2026 – May 2026

- Designed an autonomous vacuum vehicle integrating a **Raspberry Pi Pico W** with **HC-SR04 ultrasonic sensors** and an **MPU6050** gyroscope for spatial awareness
- Programmed and debugged closed-loop **MicroPython** code in VS Code for obstacle avoidance and course correction
- Designed and 3D printed custom chassis in **Onshape**, iterating on the design to protect hardware from collisions
- Tuned obstacle-avoidance logic so the vehicle comes to full stop and reverses course within **3 cm** of an obstacle to maximize floor coverage for complete room cleaning

AI Dashcam Prototype – [SHAD Canada](#)

June 2025 – July 2025

- Collaborated within an **8-person** multidisciplinary team to conceptualize the system architecture and business case for an AI-powered dashcam designed to automate road damage detection and reporting
- Co-presented the team's concept and market viability to industry judges, successfully communicating the team's technical vision to win '**Best Presentation**' and '**Best Prototype**' at the SHAD program out of **10 teams**